

Metric Space

Labix

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1 Metric Spaces

1.1 Basic Definitions

Definition 1.1.1 (Metric)

Let X be a set. A metric on X is a function $d : X \times X \rightarrow \mathbb{R}$ such that the following are true.

- For all $x, y \in X$, we have $d(x, y) \geq 0$, and $d(x, y) = 0$ if and only if $x = y$.
- $d(x, y) = d(y, x)$
- $d(x, y) \leq d(x, z) + d(z, y)$

Definition 1.1.2 (Metric Space)

A metric space is an ordered pair (X, d) where X is a set and d is a metric on X .

Example 1.1.3 (The Standard Euclidean Metric on $\mathbb{R}$)

The function $d : \mathbb{R} \times \mathbb{R} \rightarrow \mathbb{R}$ defined by

$$d(x, y) = \sqrt{x^2 + y^2}$$

defines a metric on $\mathbb{R}$, called the standard Euclidean metric.

Example 1.1.4 (Exotic Metrics in $\mathbb{R}^2$)

The following functions define metrics on $\mathbb{R}^2$.

- Define the standard Euclidean metric on $\mathbb{R}^2$ by

$$d((x_1, y_1), (x_2, y_2)) = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}$$

- Define the taxicab metric on $\mathbb{R}^2$ by

$$d((x_1, y_1), (x_2, y_2)) = |x_1 - x_2| + |y_1 - y_2|$$

- Define the French railway metric on $\mathbb{R}^2$ by

$$d((x_1, y_1), (x_2, y_2)) = \begin{cases} \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2} & \text{if } x_2 y_1 = y_2 x_1 \\ \sqrt{x_1^2 + x_2^2} + \sqrt{y_1^2 + y_2^2} & \text{otherwise} \end{cases}$$

- Define the jungle river metric on $\mathbb{R}^2$ by

$$d((x_1, y_1), (x_2, y_2)) = \begin{cases} |y_1 - y_2| & \text{if } x_1 = x_2 \\ |y_1| + |x_1 - x_2| + |y_2| & \text{otherwise} \end{cases}$$

- Define the box metric on $\mathbb{R}^2$ by

$$d((x_1, y_1), (x_2, y_2)) = \max\{|x_1 - x_2|, |y_1 - y_2|\}$$

Example 1.1.5 (The Discrete Metric)

Let X be a set. Define the discrete metric on X to be the function $d : X \times X \rightarrow \mathbb{R}$ given by

$$d(x, y) = \begin{cases} 0 & \text{if } x = y \\ 1 & \text{otherwise} \end{cases}$$

1.2 New Metrics from Old

Lemma 1.2.1 (Metric Subspace)

Let (X, d) be a metric space. Let $U \subseteq X$ be a subset. Then $(U, d|_{U \times U})$ is also a metric space

Proof

$d|_{U \times U}$ has the same properties as d . ■

Definition 1.2.2 (The p Metric) Let (X, d_X) and (Y, d_Y) be metric spaces. Let $p \in [1, \infty)$. Define the p metric on $X \times Y$ to be the map $d_p : (X \times Y) \times (X \times Y) \rightarrow \mathbb{R}$ given by

$$d_p((x_1, y_1), (x_2, y_2)) = (d_X(x_1, x_2)^p + d_Y(y_1, y_2)^p)^{1/p}$$

Proposition 1.2.3 Let (X, d_X) and (Y, d_Y) be metric spaces. Let $p \in [1, \infty)$. Then d_p is a metric on $X \times Y$.

Example 1.2.4

Let $n \in \mathbb{N}$. Then the function $d : \mathbb{R}^n \times \mathbb{R}^n \rightarrow \mathbb{R}$ defined by

$$d((x_1, \dots, x_n), (y_1, \dots, y_n)) = \sqrt{\sum_{i=1}^n (x_i - y_i)^2}$$

is a metric in $\mathbb{R}^n$, called the standard Euclidean metric of $\mathbb{R}^n$.

1.3 Size and Distance of Subsets

Definition 1.3.1 (Open Balls)

Let (X, d) be a metric space. Let $x \in X$. Let $r > 0$. Define the open ball of radius r at x to be

$$B_r(x) = \{y \in X \mid d(x, y) < r\}$$

Definition 1.3.2 (Bounded Subsets)

Let X be a metric space. Let $U \subseteq X$ be a subset. We say that U is bounded if there exists $x \in X$ and $r > 0$ such that $U \subseteq B_r(x)$.

Definition 1.3.3 (Distance between two Subsets)

Let X be a metric space. Let $U, V \subseteq X$ be subsets. Define the distance between U and V to be

$$\text{dist}(U, V) = \inf\{d(u, v) \mid u \in U, v \in V\}$$

1.4 Open and Closed Subsets

Definition 1.4.1 (Open and Closed Sets)

Let X be a metric space. Let $U \subseteq X$.

- We say that U is an open subset of X if for all $x \in U$, there exists $r > 0$ such that $B_r(x) \subseteq U$.
- We say that U is closed if $X \setminus U$ is open.

Example 1.4.2

Consider the following subsets of $\mathbb{R}$.

- $(0, 1)$ is open in $\mathbb{R}$.
- $[0, 1]$ is closed in $\mathbb{R}$.
- $(0, 1]$ and $[0, 1)$ are neither open nor closed in $\mathbb{R}$.

Example 1.4.3

Let $[0, 1)$ inherit a metric subspace structure from $\mathbb{R}$.

- $[1/2, 1)$ is closed in $[0, 1)$.
- $[0, 1)$ is both open and closed in $[0, 1)$.

Lemma 1.4.4

Let X be a metric space. Let $x \in X$. Let $r > 0$. Then $B_r(x)$ is an open subset of X .

Proof Let $B_r(a)$ be our open ball. Let $x \in B_r(a)$. Then

$$B_{(r-d(x,a))/2}(x) \subseteq B_r(a)$$

thus we are done. ■

Proposition 1.4.5

Let X be a metric space. Then the following are true.

- Let $\{U_i \mid i \in I\}$ be a collection of open subsets of X . Then $\bigcup_{i \in I} U_i$ is an open subset of X .
- Let $U_1, \dots, U_n$ be open subsets of X . Then $\bigcap_{i=1}^n U_i$ is an open subset of X .
- Let $\{U_i \mid i \in I\}$ be a collection of closed subsets of X . Then $\bigcap_{i \in I} U_i$ is a closed subset of X .
- Let $U_1, \dots, U_n$ be closed subsets of X . Then $\bigcup_{i=1}^n U_i$ is a closed subset of X .

Proof

Suppose that $\{U_i \mid i \in I\}$ is a collection of open subsets of X . Let $x \in \bigcup_{i \in I} U_i$. Then there exists $i \in I$ such that $x \in U_i$. Since U_i is open, there exists $r > 0$ such that $B_r(x) \subseteq U_i$. Then $B_r(x) \subseteq \bigcup_{i \in I} U_i$ and so we are done.

Suppose that $U_1, \dots, U_n$ are open subsets of X . Let $x \in \bigcap_{i=1}^n U_i$. Then for $1 \leq i \leq n$, there exists $r_i > 0$ such that $B_{r_i}(x) \subseteq U_i$. Let $r = \min\{r_1, \dots, r_n\}$. Then for $1 \leq i \leq n$, we have $B_r(x) \subseteq B_{r_i}(x) \subseteq U_i$. Hence $B_r(x) \subseteq \bigcap_{i=1}^n U_i$. Thus $\bigcap_{i=1}^n U_i$ is an open subset.

Suppose that $\{U_i \mid i \in I\}$ is a collection of closed subsets of X . Then $\{X \setminus U_i \mid i \in I\}$ is a collection of open subsets of X . Hence

$$\bigcup_{i \in I} (X \setminus U_i) = X \setminus \bigcap_{i \in I} U_i$$

is an open subset of X . Hence $\bigcap_{i \in I} U_i$ is a closed subsets of X .

Suppose that $U_1, \dots, U_n$ are closed subsets of X . Then $X \setminus U_i$ for $1 \leq i \leq n$ are open subsets. Hence

$$\bigcap_{i=1}^n (X \setminus U_i) = X \setminus \bigcup_{i=1}^n U_i$$

is an open subset of X by the above. Hence $\bigcup_{i=1}^n U_i$ is a closed subset of X . ■

1.5 Distance Preserving Maps

Definition 1.5.1 (Distance Preserving Maps)

Let (X, d_X) and (Y, d_Y) be metric spaces. Let $f : X \rightarrow Y$ be a function. We say that f is distance preserving if for any $p, q \in X$, we have

$$d_Y(f(p), f(q)) = d_X(p, q)$$

Definition 1.5.2 (Isometries) Let (X, d_1) and (X, d_2) be metric spaces. Let $f : X \rightarrow Y$ be a distance preserving map. We say that f is an isometry if there exists a distance preserving map $g : Y \rightarrow X$ such that

$$g \circ f = \text{id}_X \quad \text{and} \quad f \circ g = \text{id}_Y$$

It is also common to use the term isometry to mean a distance preserving function and refer to our version of isometry as bijective isometries.

Lemma 1.5.3 Let (X, d_1) and (X, d_2) be metric spaces. Let $f : X \rightarrow Y$ be a distance preserving map. Then f is an isometry if and only if f is bijective.

Proof

Clearly, if f is an isometry then f has an inverse function. Conversely, suppose that f is bijective. Then we have

$$d_X(f^{-1}(p), f^{-1}(q)) = d_Y(f(f^{-1}(p)), f(f^{-1}(q))) = d_Y(p, q)$$

for any $p, q \in Y$ since f is distance preserving. Hence f^{-1} is distance preserving. ■

Definition 1.5.4 (The Set of Isometries of a Metric Space)

Let (X, d) be a metric space. Define the set of isometries of X to be

$$\text{Isom}(X, d) = \{f : X \rightarrow X \mid f \text{ is an isometry} \}$$

2 Limits and Continuity

2.1 Sequences

Definition 2.1.1 (Sequences)

Let X be a metric space. A sequence in X is a collection of elements $a_n \in X$ for each $n \in \mathbb{N}$. We denote the sequence by $(a_n)_{n \in \mathbb{N}}$.

Definition 2.1.2 (Convergence)

Let (X, d) be a metric space. Let $(x_n)_{n \in \mathbb{N}}$ be a sequence in X . Let $x \in X$. We say that the sequence converges to the limit x if for all $\varepsilon > 0$, there exists $N \in \mathbb{N}$ such that $d(x_n, x) < \varepsilon$ for all $n > N$.

Lemma 2.1.3

Let X be a metric space. Let $(x_n)_{n \in \mathbb{N}}$ be a sequence in X . Suppose that $x, y \in X$ are the limits of the sequence, then $x = y$.

Proof

Let $\varepsilon > 0$. Since $(x_n)_{n \in \mathbb{N}}$ converges to x and y , there exists $N_1, N_2 \in \mathbb{N}$ such that $d(x_n, x) < \frac{\varepsilon}{2}$ and $d(x_m, y) < \frac{\varepsilon}{2}$ for all $n > N_1$ and $m > N_2$ respectively. Let $N = \max\{N_1, N_2\}$. Then we have $d(x_n, x) < \frac{\varepsilon}{2}$ and $d(x_n, y) < \frac{\varepsilon}{2}$ for all $n \in \mathbb{N}$. By the triangle inequality, we have

$$d(x, y) < d(x_n, x) + d(x_n, y) = \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon$$

Since $0 \leq d(x, y) < \varepsilon$ for all $\varepsilon > 0$, we deduce that $d(x, y) = 0$. Hence $x = y$. ■

Proposition 2.1.4

Let X be a metric space. Let $U \subseteq X$. Then U is closed if and only if for all sequence $(x_n)_{n \in \mathbb{N}}$ in U that converges in $x \in X$, we have $x \in U$.

Proof Suppose that U is closed. Then $X \setminus U$ is open by definition. Let $\{x_n\} \subset U$ converge to $x \notin U$. Then $x \in X \setminus U$. By definition of convergence, for every $\epsilon > 0$, there exists $N \in \mathbb{N}$ such that $x_n \in B_\epsilon(x)$ for $n > N$. But since $X \setminus U$ is open, there should be some ϵ such that $B_\epsilon(x) \subset X \setminus U$. In this case, we would have $x_n \in B_\epsilon(x) \subset X \setminus U$ which is a contradiction.

Suppose that the right hand side is true. Suppose for a contradiction that $X \setminus U$ is not open. Then for every $\epsilon > 0$, $B_\epsilon(x)$ is not a subset of $X \setminus U$ for some $x \in X \setminus U$. Let $y_k \in B_{1/k}(x)$ but $y_k \notin X \setminus U$. Then $y_k \in U$ and $y_k \rightarrow x \in X \setminus U$, a contradiction. ■

Proposition 2.1.5 Let X, Y be metrics spaces. Let $((x_n, y_n))_{n \in \mathbb{N}}$ be a sequence in $X \times Y$. Let $(x, y) \in X \times Y$. Then $((x_n, y_n))_{n \in \mathbb{N}}$ converges to (x, y) if and only if $(x_n)_{n \in \mathbb{N}}$ converges to x and $(y_n)_{n \in \mathbb{N}}$ converges to y .

Definition 2.1.6 (Cauchy Sequence) We say that $\{x_n\} \subset (X, d)$ is a Cauchy sequence if for every $\epsilon > 0$, there exists some N such that $d(x_n, x_m) < \epsilon$ for all $n, m > N$.

Proposition 2.1.7 Every convergent sequence is Cauchy.

Proof Let $(x_n)_{n \in \mathbb{N}}$ be a convergent sequence in a metric space X . Let $\epsilon > 0$, then from convergence we have that for $d(x_n, x) < \frac{\epsilon}{2}$ for all $n > N$ for some $N \in \mathbb{N}$. Then choosing the same N , we have that

$$d(x_n, x_m) \leq d(x_n, x) + d(x, x_m) < \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon$$

thus we are done. ■

We now give the definition of a complete space in terms of Cauchy sequences.

2.2 Limits of Functions

Definition 2.2.1 (Accumulation Points)

Let M be a metric space. Let $x \in M$. We say that x is an accumulation point of M if for all $r \in \mathbb{R}^+$, we have

$$B_r(x) \setminus \{x\} \neq \emptyset$$

Definition 2.2.2 (Limits)

Let $(X, d_X), (Y, d_Y)$ be metric spaces. Let $f : X \rightarrow Y$ be a function. Let $p \in X$ be an accumulation point. We say that $L \in Y$ is the limit of f at p if for all $\varepsilon > 0$, there exists $\delta > 0$ such that for all $x \in M$ such that $0 < d_X(x, p) < \delta$, we have

$$d_Y(f(x), L) < \varepsilon$$

In this case, we write $L = \lim_{x \rightarrow p} f(x)$.

2.3 Continuous Functions

Definition 2.3.1 (Continuity)

Let $(X, d_X), (Y, d_Y)$ be metric spaces. Let $f : X \rightarrow Y$ be a function. Let $p \in X$. We say that f is continuous at p if for all $\varepsilon > 0$, there exists $\delta > 0$ such that for any $x \in X$ such that $0 < d_X(x, p) < \delta$, we have $d_Y(f(x), f(p)) < \varepsilon$.

Proposition 2.3.2

Let X, Y be metric spaces. Let $f : X \rightarrow Y$ be a function. Let $p \in X$ be an accumulation point. Then the following are equivalent.

- f is continuous at p .
- For every sequence $(x_n)_{n \in \mathbb{N}} \subseteq X$ such that $(x_n)_{n \in \mathbb{N}} \rightarrow x$, we have $(f(x_n))_{n \in \mathbb{N}} \rightarrow f(p)$.
- $\lim_{x \rightarrow p} f(x) = f(p)$.

Proof ■

Proposition 2.3.3

Let X, Y be metric spaces. Let $f : X \rightarrow Y$ be a function. Then the following are equivalent.

- f is continuous.
- For all $y \in Y$ and $r > 0$, we have that $f^{-1}(B_r(y))$ is open in X .
- For all open subsets $V \subseteq Y$, we have that $f^{-1}(V)$ is open in X .

Proof

Suppose that f is continuous. Let $r > 0$. By continuity, there exists $\delta > 0$ such that $f(B_\delta(p)) \subset B_r(f(p))$. This implies that $B_\delta(p) \subset f^{-1}(B_r(f(p)))$. Thus $f^{-1}(B_r(f(p)))$ is open.

Now suppose that for all $y \in Y$ and $r > 0$, we have $f^{-1}(B_r(y))$ is open in X . Let $V \subseteq Y$ be an open subset. Let $x \in f^{-1}(V)$. Then there exists $r > 0$ such that $B_r(f(x)) \subseteq V$. This means that $x \in f^{-1}(B_r(f(x))) \subseteq f^{-1}(V)$. Since $f^{-1}(B_r(f(x)))$ is open, there exists $s > 0$ such that

$$B_s(x) \subseteq f^{-1}(B_r(f(x))) \subseteq f^{-1}(V)$$

Since this is true for all $x \in f^{-1}(V)$, we conclude that $f^{-1}(V)$ is open.

Now suppose that $\Omega \subset V$ is open imply $f^{-1}(\Omega)$ is open. Let $p \in \Omega$. Then there exists $\epsilon > 0$ such that $B_\epsilon(f(p)) \subset \Omega$. By assumption, we must have $f^{-1}(B_\epsilon(f(p)))$ is open. The fact that this is open means there exists $\delta > 0$ such that $B_\delta(p) \subset f^{-1}(B_\epsilon(f(p)))$. Then we have

$$f(B_\delta(p)) \subset B_\epsilon(f(p))$$

and we are done. ■

Example 2.3.4

Let X be a metric space. Let $y \in X$. Prove that the map $d(-, y) : X \rightarrow \mathbb{R}$ is continuous.

Answer

Let $(a, b) \subseteq \mathbb{R}$. If the inverse image is empty, then we are done. So suppose that $z \in \{x \in X \mid a < d(x, y) < b\}$. Choose $0 < r < b - d(z, y)$. Then for any $w \in B_r(z)$, we have

$$d(w, y) \leq d(w, z) + d(z, y) < r + d(z, y) < b - d(z, y) + d(z, y) = b$$

Hence $B_r(z) \subseteq \{x \in X \mid a < d(x, y) < b\}$ and so the latter is an open set. By the above proposition we conclude. ⊛

Lemma 2.3.5

Let (X, d_1) and (Y, d_2) be metric spaces. Let $f : X \rightarrow Y$ be a distance preserving map. Then f is continuous.

Proof Let $p \in X$. Let $\varepsilon > 0$. Choose $\delta = \varepsilon$, then for all $x \in X$ such that $0 < d_X(x, p) < \delta$, we have $d_Y(f(x), f(p)) = d_X(x, p) < \delta = \varepsilon$ so that f is continuous. ■

Proposition 2.3.6

Let X, Y, Z be metric spaces. Let $f : X \rightarrow Y$ and $g : Y \rightarrow Z$ be functions. If f and g are continuous, then $g \circ f$ is continuous.

2.4 Uniform Continuous Functions

Definition 2.4.1 (Uniformly Continuous) A map $f : X \rightarrow Y$ is uniformly continuous if for every $\epsilon > 0$, there exists $\delta > 0$ such that

$$d_X(x, y) < \delta \implies d_Y(f(x), f(y)) < \epsilon$$

for any $x, y \in X$.

Lemma 2.4.2 If $f : X \rightarrow Y$ is uniformly continuous then it is continuous.

2.5 Lipschitz Continuous Functions

Definition 2.5.1 (Lipschitz Continuous) Let (X, d_X) and (Y, d_Y) be metric spaces and suppose that $f : X \rightarrow Y$. We say that f is a Lipschitz map if there is a constant $K \geq 0$ such that

$$d_Y(f(x), f(y)) \leq K d_X(x, y)$$

for all x, y in X .

Lemma 2.5.2 If $f : X \rightarrow Y$ is Lipschitz continuous then it is uniformly continuous.

Definition 2.5.3 (Contraction Maps)

Let X be a metric space. Let $f : X \rightarrow X$ be a function. We say that f is a contraction map if f is Lipschitz continuous with $K \in [0, 1)$.

3 Convergence in Metric Spaces

3.1 Complete Metric Spaces

Definition 3.1.1 (Complete Spaces)

Let X be a metric space. We say that X is complete if every Cauchy sequence in X converges.

Proposition 3.1.2

Let X be a metric space. Let $Y \subseteq X$ be a subset. Then the following are true.

- If X is complete and Y is closed, then Y is complete.
- If Y is complete, then Y is closed.

Proof

Suppose that X is a complete metric space and Y is closed. Let $(y_n)_{n \in \mathbb{N}}$ be a Cauchy sequence in Y . Since it is also a sequence in X , completeness of X implies that it converges to some $x \in X$. But since Y is closed, we must have $x \in Y$. Hence Y is complete.

Now suppose that Y is complete. Then for any convergent sequence $(y_n)_{n \in \mathbb{N}}$ in Y , it is Cauchy and hence converges to some $Y \in Y$. Thus Y is closed. ■

Theorem 3.1.3 (Cantor's Intersection Theorem) Let X be a complete metric space. Let $S_1 \supseteq S_2 \supseteq \dots$ form a nested sequence of non-empty closed sets in X with the property that $\text{diam}(S_n) \rightarrow 0$ as $n \rightarrow \infty$. Then

$$\bigcap_{n=1}^{\infty} S_n \neq \emptyset$$

Proof For each $N \in \mathbb{N}$, choose $x_N \in S_N$. Then for all $n > N$, $x_n \in S_N$. Thus for $n, m > N$, we have that $d(x_n, x_m) \leq \text{diam}(S_N)$. It follows that $(x_n)_{n \in \mathbb{N}}$ is Cauchy. Thus $x_n \rightarrow x$ for some $x \in X$. Since each S_n is closed and $x_n \in S_N$ for all $n > N$, we must have that $x \in S_n$ for each n . Thus $x \in \bigcap_{k=1}^{\infty} S_n$ is nonempty. ■

Below are a few examples of complete spaces.

Example 3.1.4

$\mathbb{R}^n$ is a complete metric space.

Proof

Let $(x_k)_{k \in \mathbb{N}}$ be a Cauchy sequence in $\mathbb{R}^n$. Denote the i th component of x_k by $x_{k,i}$. Then for every $\epsilon > 0$, there exists N such that

$$\|x_k - x_m\| = \left(\sum_{i=1}^n |x_{k,i} - x_{m,i}|^2 \right)^{\frac{1}{2}} < \epsilon$$

for $k, m > N$. In particular, we have that each individual

$$|x_{k,i} - x_{m,i}| < \epsilon$$

for $m, n > N$. Thus $(x_{k,i})_{k \in \mathbb{N}}$ is a Cauchy sequence in $\mathbb{R}$. But we know that Cauchy sequences in $\mathbb{R}$ converges, thus $(x_{k,i})_{k \in \mathbb{N}}$ converges to $x_i \in \mathbb{R}$. Now define $x = (x_1, \dots, x_n)$, then

$$\|x_k - x\| = \left(|x_{k,i} - x_i|^2 \right)^{\frac{1}{2}} < n\epsilon$$

by convergence of each individual component. Thus $(x_n)_{n \in \mathbb{N}}$ is a convergent sequence. ■

3.2 Completion

The goal of this section is to attempt to complete a metric space by adding in the missing limits of a metric space.

Definition 3.2.1 (Space of Bounded Real Functions)

Let X be a metric space. Define the set of bounded real functions of X to be

$$B(X) = \{f : X \rightarrow \mathbb{R} \mid |f| \leq M \text{ for some } M \in \mathbb{R}\}$$

Definition 3.2.2 (The Supremum Metric)

Let X be a metric space. Define the supremum metric on $B(X)$ by the formula

$$\|f - g\|_\infty = \sup_{x \in X} |f(x) - g(x)|$$

Proposition 3.2.3

Let X be a metric space. Then the metric space $B(X)$ whose metric induced by the supremum norm is complete.

Proof Let $(f_n)_{n \in \mathbb{N}}$ be a Cauchy sequence in $B(X)$. Then for every $\epsilon > 0$, there exists N such that

$$\|f_n - f_m\|_\infty = \sup_{x \in X} |f_n(x) - f_m(x)| < \epsilon$$

for all $n, m > N$. In particular, for each $x \in X$, the property of supremum implies that $|f_n(x) - f_m(x)| < \epsilon$ for $n, m > N$. Thus $(f_n(x))_{n \in \mathbb{N}} \subset \mathbb{R}$ is Cauchy for each x . Since $\mathbb{R}$ is complete, $(f_n(x))_{n \in \mathbb{N}}$ converges for each $x \in X$.

Now define the function $f : X \rightarrow \mathbb{R}$ by

$$f(x) = \lim_{n \rightarrow \infty} f_n(x)$$

Then fix $\epsilon > 0$, we have that

$$|f_n(x) - f(x)| < \epsilon$$

for all $n > N$ by letting $m \rightarrow \infty$ from the fact that $|f_n(x) - f_m(x)| < \epsilon$. This N does not depend on x . Fix $\epsilon = 1$, then there exists $N_1 \in \mathbb{N}$ such that

$$\begin{aligned} |f(x) - f_n(x)| &\leq |f(x) - f_{N_1}(x)| \\ &\leq 1 + |f_{N_1}(x)| \end{aligned}$$

for all $x \in X$ and $n > N_1$ thus f is bounded. This means that $f \in B(X)$ and that $\|f_n - f\|_\infty < \epsilon$ for all $n > N$. ■

Proposition 3.2.4

Let X be a metric space. Then there exists a function $f : X \rightarrow B(X)$ such that map $f : X \rightarrow \text{im}(f)$ is an isometry.

3.3 The Contraction Mapping Theorem

Theorem 3.3.1 (Contraction Mapping Theorem) Let X be a nonempty complete metric space and suppose that $f : X \rightarrow X$ is a contraction. Then f has a unique fixed point, meaning there is a unique $x \in X$ such that $f(x) = x$.

Proof Let $x_0 \in X$ and define a sequence by $x_{n+1} = f(x_n)$ for $n \in \mathbb{N}$. Then we have that

$$d(x_{n+1}, x_n) \leq K d(x_n, x_{n-1}) \leq \cdots \leq K^n d(x_1, x_0)$$

Then for any $k > n$, we have that

$$\begin{aligned} d(x_k, x_n) &\leq \sum_{i=n}^{k-1} d(x_{i+1}, x_i) \\ &\leq \sum_{i=n}^{k-1} K^i d(x_1, x_0) \\ &\leq \frac{K^n}{1-K} d(x_1, x_0) \end{aligned}$$

This is Cauchy since we can choose $\epsilon > 0$ such that $\frac{K^n}{1-K} < \epsilon$. Since X is complete, we have that $x_n \rightarrow x$ for some $x \in X$. Since f is continuous we have that $f(x_n) \rightarrow f(x)$. Now taking limits on

$$x_{n+1} = f(x_n)$$

we have that $x = f(x)$.

To prove uniqueness, note that if $f(x) = x$ and $f(y) = y$, then

$$d(x, y) = d(f(x), f(y)) \leq K d(x, y)$$

which implies that $(1 - K)d(x, y) = 0$. Thus $x = y$. ■

Another name for this theorem would be Banach's Fixed Point Theorem.

Theorem 3.3.2 (Picard-Lindelof Theorem) Let $f : \mathbb{R}^n \rightarrow \mathbb{R}^n$ be Lipschitz continuous with

$$|f(x) - f(y)| \leq L|x - y|$$

where $x, y \in \mathbb{R}^n$. Then for any $x_0 \in \mathbb{R}^n$, the differential equation

$$\frac{dx}{dt} = f(x)$$

with initial condition $x(0) = x_0$ has a unique solution on $[-t, t]$ for any $Lt < 1$.

3.4 Cantor's Theorem

Theorem 3.4.1 If X is a complete metric space and $\{F_n | n \in \mathbb{N}\}$ is a collection of open dense subsets of X , then

$$F = \bigcap_{k=1}^{\infty} F_k$$

is dense in X . Equivalently, if $\{G_n | n \in \mathbb{N}\}$ is a collection of nowhere dense subsets of a nonempty complete metric space X , then

$$\bigcup_{k=1}^{\infty} F_k \neq X$$

Lemma 3.4.2 The Cantor set is uncountable.

4 Paths in Metric Spaces

4.1 Basic Properties of Paths

Definition 4.1.1 (Paths in a Metric Space)

Let X be a metric space. A path in X is a continuous function $f : [a, b] \rightarrow X$.

Definition 4.1.2 (Different Types of Paths)

Let X be a metric space. Let $f : [a, b] \rightarrow X$ be a path in X .

- We say that r is simple if $f(t_1) = f(t_2)$ implies $t_1 = t_2$ for all $t_1, t_2 \in (a, b)$.
- We say that r is closed if $f(a) = f(b)$.

Definition 4.1.3 (Reparametrization)

Let X be a metric space. Let $f : [a, b] \rightarrow X$ and $g : [c, d] \rightarrow X$ be paths in X . We say that g is a reparametrization of f if there exists a continuous function $\rho : [c, d] \rightarrow [a, b]$ such that $g = f \circ \rho$.

Definition 4.1.4 (Orientation Preserving Reparametrizations) Let X be a metric space. Let $f : [a, b] \rightarrow X$ be a path in X . Let $g : [c, d] \rightarrow X$ be a reparametrization of f via the continuous function $\rho : [c, d] \rightarrow [a, b]$. We say that g is an orientation preserving reparametrization if $\rho'(t) > 0$ for all $t \in [c, d]$.

4.2 Geodesics

Definition 4.2.1 (Geodesics)

Let X be a metric space. Let $I \subseteq \mathbb{R}$ be an interval. Let $\gamma : I \rightarrow X$ be a path. We say that γ is a geodesic if γ is distance preserving.

Definition 4.2.2 (Geodesic Metric Spaces) Let (X, d) be a metric space. We say that X is geodesic if for all $a, b \in X$, there exists a geodesic $\gamma : I \rightarrow X$ from a to b .